Bhartiya Antariksh Hackathon’25

PS: Identifying Halo CME events based on particle data from SWIS-ASPEX payload onboard Aditya-L1

WHAT WAS GIVEN ON WEBSITE:

Overview

Identifying Halo CME Events Based on Particle Data from SWIS-ASPEX Payload onboard Aditya-L1

Our Sun drives the particle flux in the interplanetary medium and also affects our near earth environment. A sudden change in these particle flux can result in significant disturbance in the upper atmosphere and can result in major loss to the space assets. An early warning system based on particle measurement carried out at location much farther away, before these particles reach earth, can help us to take precautionary measures to prevent any catastrophic destruction.

Objective:

• To use the data from Solar Wind Ion Spectrometer (SWIS) instrument of ASPEX payload onboard Aditya-L1 mission to identify transient events such as halo Coronal Mass Ejection (CME) from the Sun.

• To analyze SWIS Level-2 data (particle flux, number density, temperature, velocity) to characterize CME signatures.

• To develop methods to process time-series of SWIS parameters and derive suitable thresholds for detecting halo CME events.

Expected Outcomes:

• Identification of threshold(s) or derived parameter(s) from SWIS data time-series that are indicative of halo CME events.

• Improved understanding of transient solar wind signatures at L1 location.

Dataset Required:

• SWIS Level-2 data (flux, number density, temperature, speed) from Aug 2024 onward, available at ISSDC.

• Halo CME event timestamps and properties from CACTUS CME database.

Suggested Tools/Technologies:

Programming Languages: Python or C

• Libraries: CDF libraries (from NASA SPDF)

• Visualization: Matplotlib, Plotly

• Signal Processing: Pandas, SciPy, NumPy

Expected Solution / Steps to be followed to achieve the objectives:

1. Identify halo CME timestamps using CACTUS CME database.

2. Extract corresponding SWIS Level-2 data for identified CME windows.

3. Analyze flux, density, temperature, and speed parameters during these windows.

4. Derive new time-series features (e.g., moving averages, gradients, combined metrics).

5. Determine statistical thresholds on derived parameters that signal CME presence. 6. Validate thresholds against confirmed CME occurrences for accuracy and robustness.

Evaluation Parameters:

• Correct identification of key signature patterns corresponding to CME events.

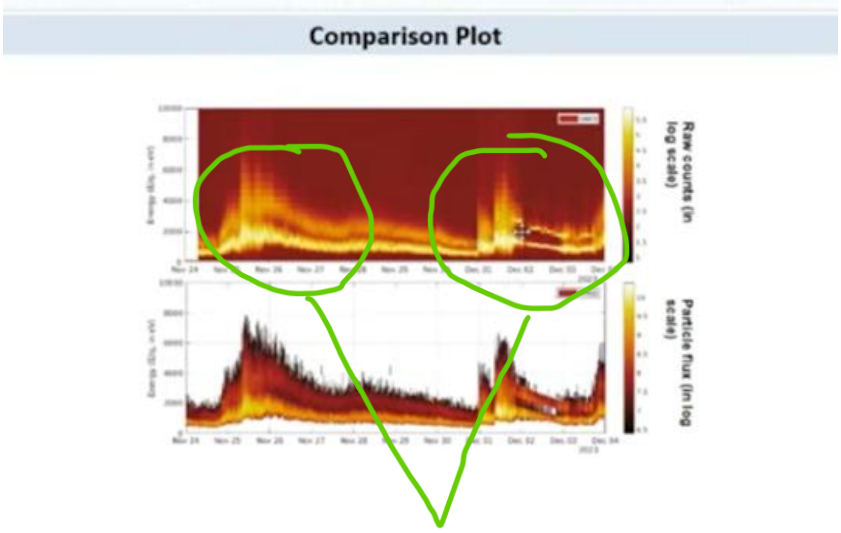
• Effectiveness of derived parameters and thresholds in distinguishing transient events.

• Accuracy and reliability of detection methodology using time-series data.

WHAT WAS TOLD IN SESSION:

Sun is quite active and constantly emits charged particles that affects the earth's magnetosphere, and eventually this connects to sun-earth's coupling and will derive the space weather events such as geomagnetic storms and everything that happens because of these coronal mass ejections and lots of events happening at the sun. But the major problem here is we don't know when this happens and how can we predict such events in future.

ISRO has sent Aditya-L1 mission.



The task is to write an algorithm to detect such high jumps in the time-series data

Objectives

The objective of the problem is to use the data from Solar Wind Ion Spectrometer (SWIS) instrument of ASPEX payload onboard Aditya-L1 mission to identify transient events such as halo Coronal Mass Ejection (CME) from the sun. The details related to the mission and the instrument can be obtained from

https://www.isro.gov.in/Aditya_L1.html and https://www.prl.res.in/ASPEX/.Briefly, the SWIS instrument measures the incoming solar wind particles (primarily proton and alpha particles) at the L1 point of Sun-Earth system. The level 2 data from SWIS contains flux of particles as well as some of the bulk parameters such as number density, temperature and velocity. The task is to use a combination of these parameters to correctiy identify halo CME events (CME events which are directed towards earth). **Note that these transient events cannot be easily identified through simple thresholds for each parameter. It'll require some processing of the time-series data of the available parameters (flux, number density, temperature and speed)**

Expected Outcomes:

The possible outcome of the problem would be a threshold or a set of thresholds on the derived parameters (obtained through processing of the time series data sets) for true identification of such events.

Dataset Required: The date range for data to be used should be from Aug. 2024 onward. The data needs to be downloaded from the ISSDC website (https://www.issdc.gov.in/). Halo CME catalog can be accessed from CACTUS CME database (https://www.sidc.be/cactus/catalog.php)

Suggested Tools/Technologies: The data is made avallable in Common Data Format (CDF). These can.be read using CDF libraries available in C and Python from NASA'S SPDF website (https://cdf.gsfc.nasa.gov/).

Expected Solution / Steps to be followed to achieve the objectives:

The steps to be followed include first identifying SWIS datasets for halo CME events based on the catalogue. The segregated data sets need to be then evaluated for specific changes during the transient events in the parameters available in the SWIS data. The next task would-then require deriving new variables (a function of parameters in the SWIS data) which would be a good indicator of transient events as it passes through the L1 point. To obtain a. suitable threshold, one would require to study pattern in the time series data of these new variables.

Some videos:

1. Aditya L1 | India's First Sun Mission Launched! | ISRO | Dhruv Rathee

🌞 **Aditya-L1 Mission: Summary**

🛰️ **What is Aditya-L1?**

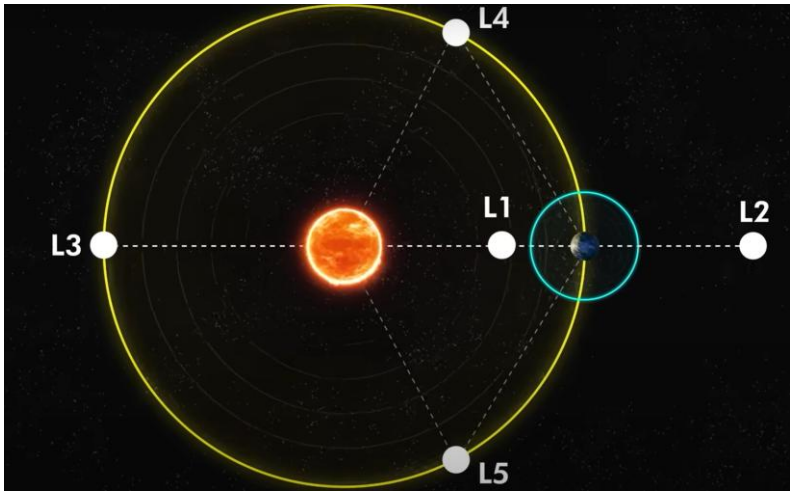
- **India's first solar observatory mission** by ISRO.
- Unlike Chandrayaan-3 (which landed on the Moon), **Aditya-L1 will not land on the Sun**—it will **observe the Sun from a distance**.



🌌 **Mission Details**

- **Orbiting Point**: Aditya-L1 will travel **1.5 million km from Earth** to orbit around **Lagrange Point L1** in a **halo orbit**.
- **Timeframe**:
  - Takes **~4 months** to reach L1.
  - Will observe the Sun for **~5 years** from that point.
- Acts as a **space observatory**, not just a spacecraft.

⚖️ **What is a Lagrange Point (L1)?**



- Points in space where the **gravitational pull** of two massive bodies (e.g., Sun and Earth) **balances the centrifugal force** of a smaller object (like a spacecraft).
- **Benefits of L1**:
    1. **Stability** – Requires little fuel to maintain position.
    2. **Continuous Observation** – Always faces the Sun and Earth without being eclipsed.
- **Other Missions** at L1: NASA-ESA's **SOHO** solar observatory.
- **James Webb Space Telescope** is at **L2**, for deep space imaging with the Sun blocked by Earth.



☀️ **The Structure of the Sun**

1. **Core**:
    - Site of **nuclear fusion**.
    - Temp: **~15 million °C**.
2. **Radiative Zone**:
    - Transfers energy via radiation.
    - Covers **70%** of Sun's radius.
3. **Convective Zone**:
    - Transfers energy via convection.
    - Covers remaining **30%**.
4. **Photosphere**:
    - The "surface" of the Sun.
    - Temp: **~5,500 °C**.
5. **Chromosphere**:
    - Above the photosphere.
    - Temp: **6,000 to 20,000 °C**.
6. **Transition Region**
7. **Corona**:
    - Outermost atmosphere.
    - Temp: **~1–3 million °C**.
    - Still a mystery why it is hotter than the surface.

☢️ **What Does the Sun Emit?**

- **Electromagnetic radiation**:
    - Visible light, **UV**, **X-rays**, **gamma rays**, **infrared**, **radio waves**.
- **Solar Wind**:
    - A stream of **charged particles (electrons & protons)**.
    - Causes **auroras** (Northern Lights).
- **Solar Flares**:
    - Sudden bursts of intense radiation.
- **Coronal Mass Ejections (CMEs)**:
    - Huge expulsions of solar wind and magnetic fields.

- Can disturb Earth's magnetosphere and satellites.

🔭 **Instruments (Payloads) on Aditya-L1**

Aditya-L1 has **7 payloads** (scientific instruments):

☀️ **Studying the Sun (4 payloads):**

1. **VELC** – *Visible Emission Line Coronagraph*
   - Studies the **Corona** and **CMEs**.
2. **SUIT** – *Solar Ultraviolet Imaging Telescope*
   - Images the **Photosphere** and **Chromosphere** in UV.
3. **SOLEXS** – *Solar Low Energy X-ray Spectrometer*
   - Observes **low energy X-rays** from solar flares.
4. **HEL1OS** – *High Energy L1 Orbiting X-ray Spectrometer*
   - Observes **high energy X-rays**.

🌐 **Studying Space Environment at L1 (3 payloads):**

5. **ASPEX** – *Aditya Solar Wind Particle Experiment*
   - Measures **solar wind particle flux**, ions, and alpha particles.
6. **PAPA** – *Plasma Analyzer Package for Aditya*
   - Analyzes **plasma properties** of solar wind.
7. **MAG** – *Magnetometer*
   - Measures **magnetic fields** at L1.

🧪 **Why Study the Sun from Space?**

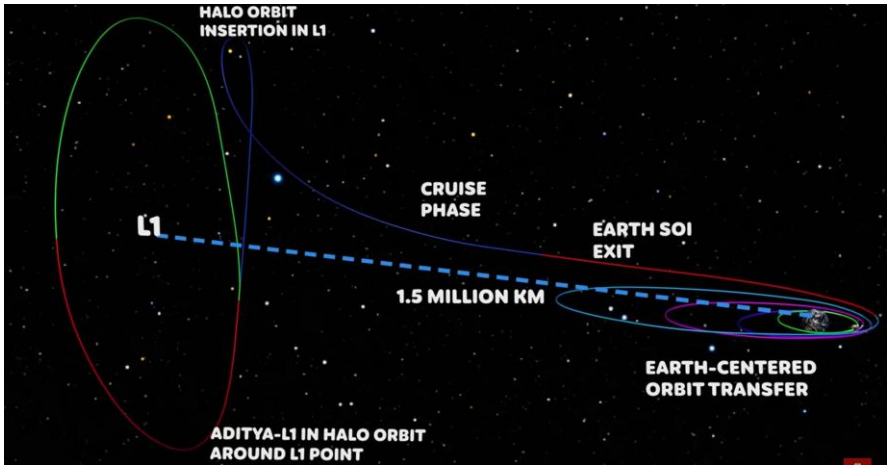
- Earth's atmosphere blocks much of the harmful radiation (like UV, X-rays), which makes **space-based instruments essential** for studying them.
- Studying solar emissions helps understand:
  - **Space weather**
  - **Satellite protection**
  - **Solar-terrestrial interactions**
  - **Mysteries like why the Corona is hotter than the surface**

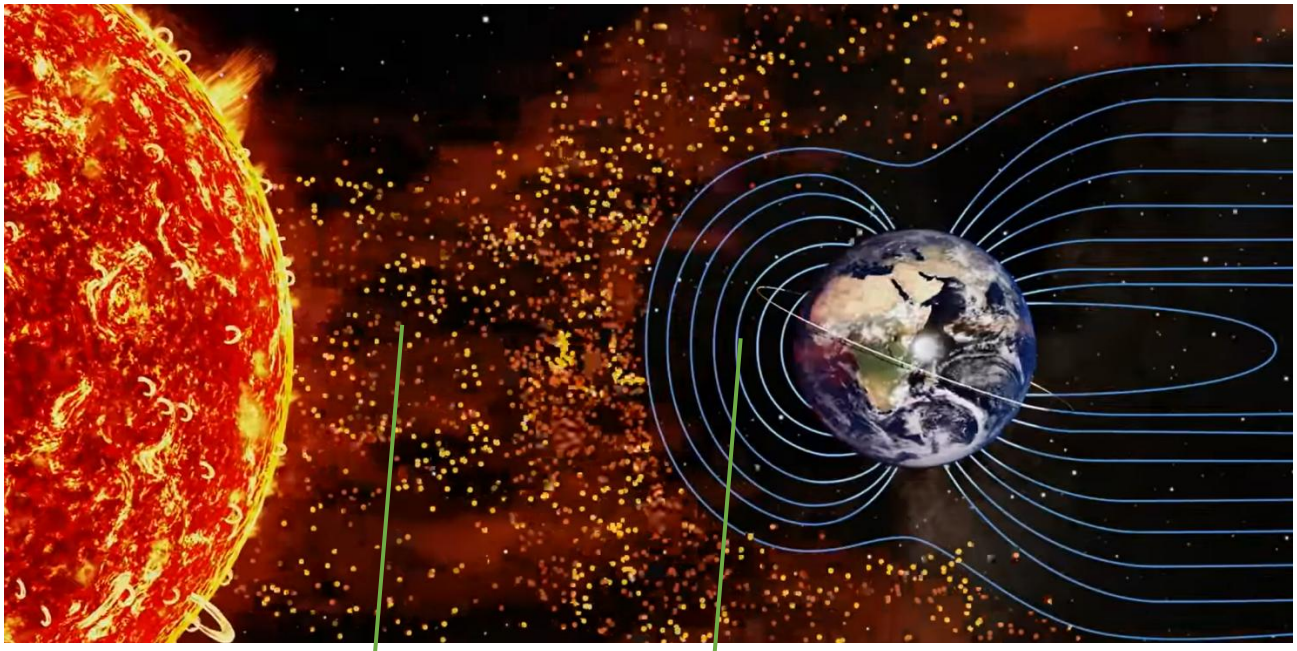
🌍 **Other Similar Missions**

- **NASA's Parker Solar Probe (2018)**:
  - Flies closer to the Sun's corona than any previous spacecraft.
- **Solar Orbiter (2020)**:
  - Aims to study Sun's poles and magnetic activity (by ESA & NASA).

2. [Aditya L1: India's Solar Mission Explained]

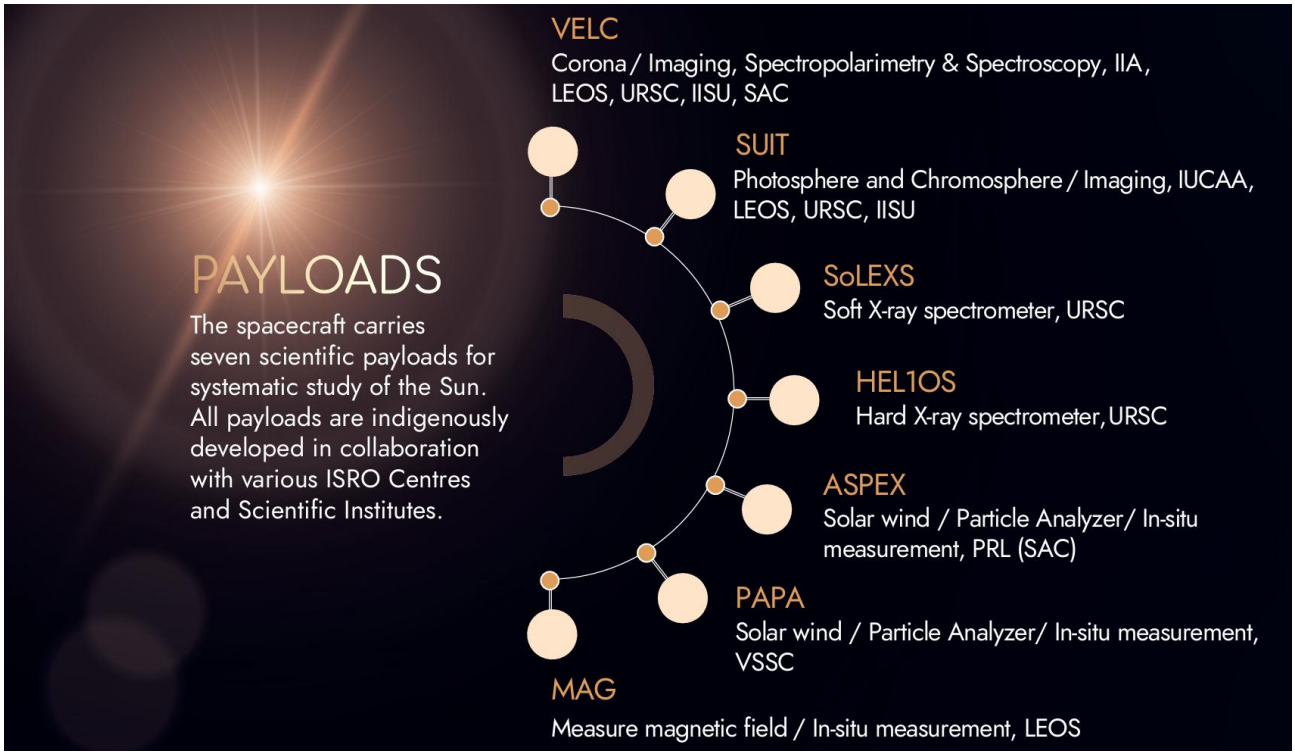
Trajectory of Aditya L1:

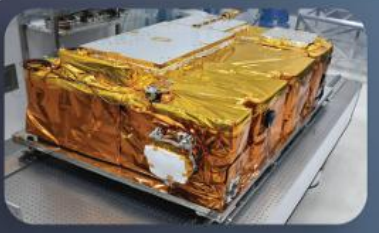




Solar storm/CME

Earth's Magnetosphere





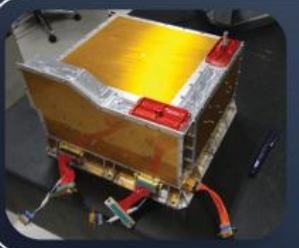
Visible Emission Line Coronagraph

VELC is the prime payload designed as a reflective coronagraph with a multi-slit spectrograph.

Solar Low Energy X-ray Spectrometer

SoLEXS is a soft X-ray spectrometer designed to measure the solar soft X-ray flux to study solar flares.





Plasma Analyser Package for Aditya

PAPA is designed to understand solar winds and its composition and, do mass analysis of solar wind ions.

High Energy L1 Orbiting X-ray Spectrometer

HEL1OS is a hard X-ray spectrometer designed to study solar flares in the high energy X-rays.



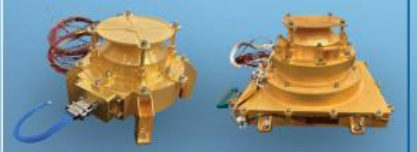


Solar Ultra-violet Imaging Telescope

SUIT is a UV telescope to image the solar disk in the near ultra-violet wavelength range.

Aditya Solar wind Particle EXperiment

ASPEX payload comprises 2 Subsystems: SWIS and STEPS



Solar Wind Ion Spectrometer is a low-energy spectrometer designed to measure the proton and alpha particles of the solar wind.



Suprathermal and Energetic Particle Spectrometer is a high-energy spectrometer designed to measure high-energy ions of the solar wind.

Magnetometer

MAG will measure the low intensity interplanetary magnetic field in space. It has two sets of Magnetic Sensors: one at the tip of a 6 meter deployable boom, and the other in the middle of the boom, 3 meters away from the spacecraft.

